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EXPERIMENT NO: 06

AIM: To implement **K-Means Clustering and Hierarchical Clustering** on a customer dataset to identify customer segments based on income and spending behavior.

1. Dataset Source

The dataset used in this experiment is the **Mall Customer Segmentation Dataset**, obtained from Kaggle.
Dataset Name: Mall Customer Segmentation Dataset
Source: Kaggle
Link:
https://www.kaggle.com/datasets/vjchoudhary7/customer-segmentation-tutorial-in-python
This dataset contains information about customers such as gender, age, annual income, and spending score which can be used for customer segmentation.

2. Dataset Description

The dataset contains information about mall customers.
Dataset Details

- **Total records:** 200
- **Total attributes:** 5

Feature	Description	Type
CustomerID	Unique Customer ID	Numerical
Gender	Gender of customer	Categorical
Age	Age of customer	Numerical
Annual Income (k$)	Annual income of the customer	Numerical
Spending Score(1 - 100)	Score assigned based on customer spending behavior	Numerical

Features used for Clustering
For clustering analysis, the following variables were selected:

- **Annual Income (k$)**
- **Spending Score (1-100)**

These features help identify groups of customers with similar purchasing behavior.

3. Mathematical Formulation

K-Means Clustering

K-Means is an unsupervised learning algorithm used to group similar data points into clusters.
The algorithm minimizes the **Within Cluster Sum of Squares (WCSS)**:

$$WCSS = \sum_{i=1}^{k} \sum_{x \in C_i} ||x - \mu_i||^2$$

Where:

- k = number of clusters
- x = data point
- C_i = cluster
- μ_i = centroid of cluster

The algorithm works by:

1. Selecting **k cluster centroids**
2. Assigning data points to nearest centroid
3. Updating centroid positions
4. Repeating until convergence

Hierarchical Clustering

Hierarchical clustering builds clusters in a **tree-like structure called a dendrogram**.

There are two approaches:

Agglomerative (Bottom-Up)

- Each data point starts as its own cluster

- Clusters are merged step-by-step

Divisive (Top-Down)

- All data points start in one cluster
- Clusters are split recursively

Distance between clusters is calculated using methods such as:

- Single linkage
- Complete linkage
- Average linkage
- Ward linkage

4. Algorithm Limitations

K-Means Limitations

- Requires pre-specifying the number of clusters (k)
- Sensitive to initial centroid selection
- Does not work well with non-spherical clusters
- Sensitive to outliers

Hierarchical Clustering Limitations

- Computationally expensive for large datasets
- Difficult to determine optimal number of clusters
- Cannot undo cluster merges once performed

5. Methodology / Workflow

Step-by-Step Workflow

1. Dataset collection from Kaggle

2. Data loading using Pandas
3. Data visualization and inspection
4. Feature selection for clustering
5. Determine optimal number of clusters using **Elbow Method**
6. Apply **K-Means Clustering**
7. Apply **Hierarchical Clustering**
8. Visualize clusters
9. Compare clustering results

6. Performance Analysis

The clustering algorithms grouped customers based on income and spending patterns.

Observations

- Customers were grouped into multiple clusters based on similar purchasing behavior.
- High-income and high-spending customers formed a distinct cluster representing potential target customers.
- Low-income and low-spending customers formed another cluster indicating low purchasing power.

Interpretation

Clustering helps businesses identify different customer segments and design targeted marketing strategies.

7. Cluster Selection (Equivalent to Hyperparameter Tuning)

The **Elbow Method** was used to determine the optimal number of clusters.

The optimal number of clusters was found where the **WCSS curve forms an elbow**, indicating diminishing returns in variance reduction.

For this dataset, **k = 5 clusters** typically provides the best segmentation.

CODE and OUTPUT:

Step 1: Import Required Libraries

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.cluster import KMeans
from sklearn.cluster import AgglomerativeClustering
from scipy.cluster.hierarchy import dendrogram, linkage
```

✓ 0.1s

Step 2: Load Dataset

```
df = pd.read_csv("Mall_Customers.csv")

print(df.head())
```

✓ 0.0s

```
   CustomerID   Genre  Age  Annual Income (k$)  Spending Score (1-100)
0           1    Male   19                  15                      39
1           2    Male   21                  15                      81
2           3  Female   20                  16                       6
3           4  Female   23                  16                      77
4           5  Female   31                  17                      40
```

Step 3: Select Features for Clustering

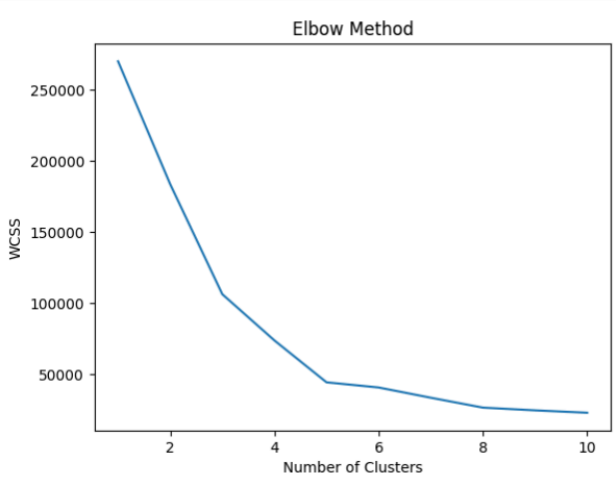
```
X = df[['Annual Income (k$)', 'Spending Score (1-100)']]
```

✓ 0.0s

Step 4: Determine Optimal Clusters using Elbow Method

```
wcss = []

for i in range(1,11):
    kmeans = KMeans(n_clusters=i, random_state=42)
    kmeans.fit(X)
    wcss.append(kmeans.inertia_)

plt.plot(range(1,11), wcss)
plt.xlabel("Number of Clusters")
plt.ylabel("WCSS")
plt.title("Elbow Method")
plt.show()
```

✓ 8.3s



PART A – K-Means Clustering

Step 5: Apply K-Means Algorithm

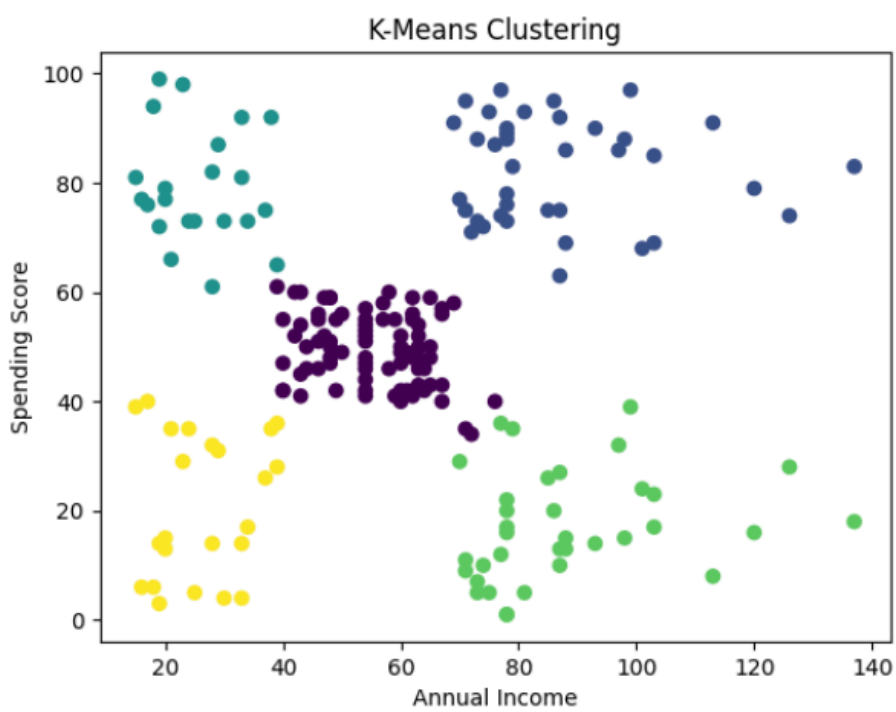
```
kmeans = KMeans(n_clusters=5, random_state=42)

clusters = kmeans.fit_predict(X)

df['Cluster'] = clusters
```

✓ 0.0s

Step 6: Visualize K-Means Clusters

```
plt.scatter(X['Annual Income (k$)'], X['Spending Score (1-100)'], c=clusters)
plt.xlabel("Annual Income")
plt.ylabel("Spending Score")
plt.title("K-Means Clustering")
plt.show()
```

✓ 0.3s

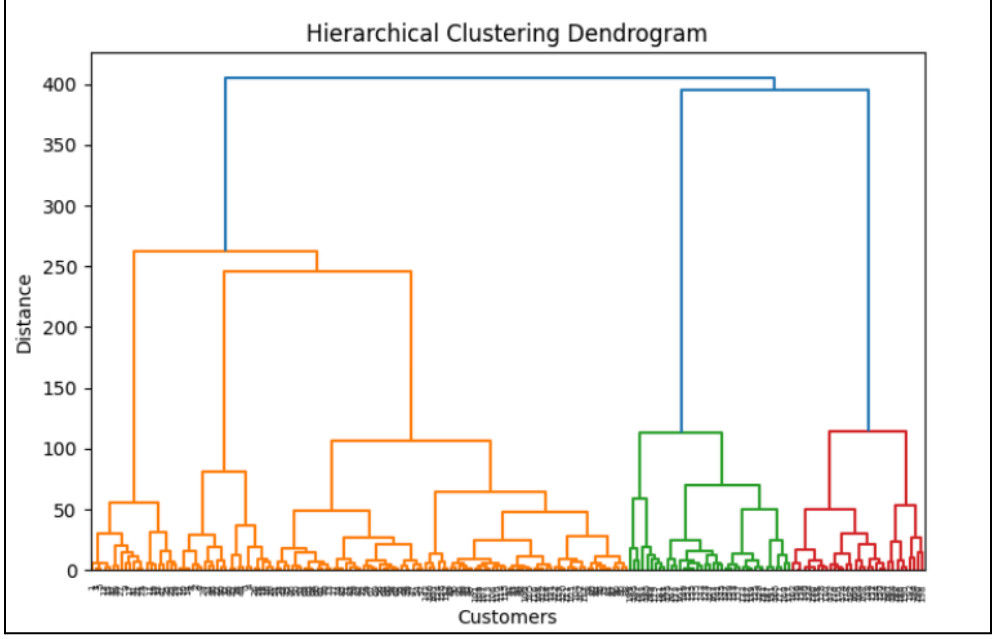


PART B – Hierarchical Clustering

Step 7: Create Dendrogram

```
Z = linkage(X, method='ward')

plt.figure(figsize=(8,5))
dendrogram(Z)
plt.title("Hierarchical Clustering Dendrogram")
plt.xlabel("Customers")
plt.ylabel("Distance")
plt.show()
```

✓ 2.0s



Step 8: Apply Hierarchical Clustering

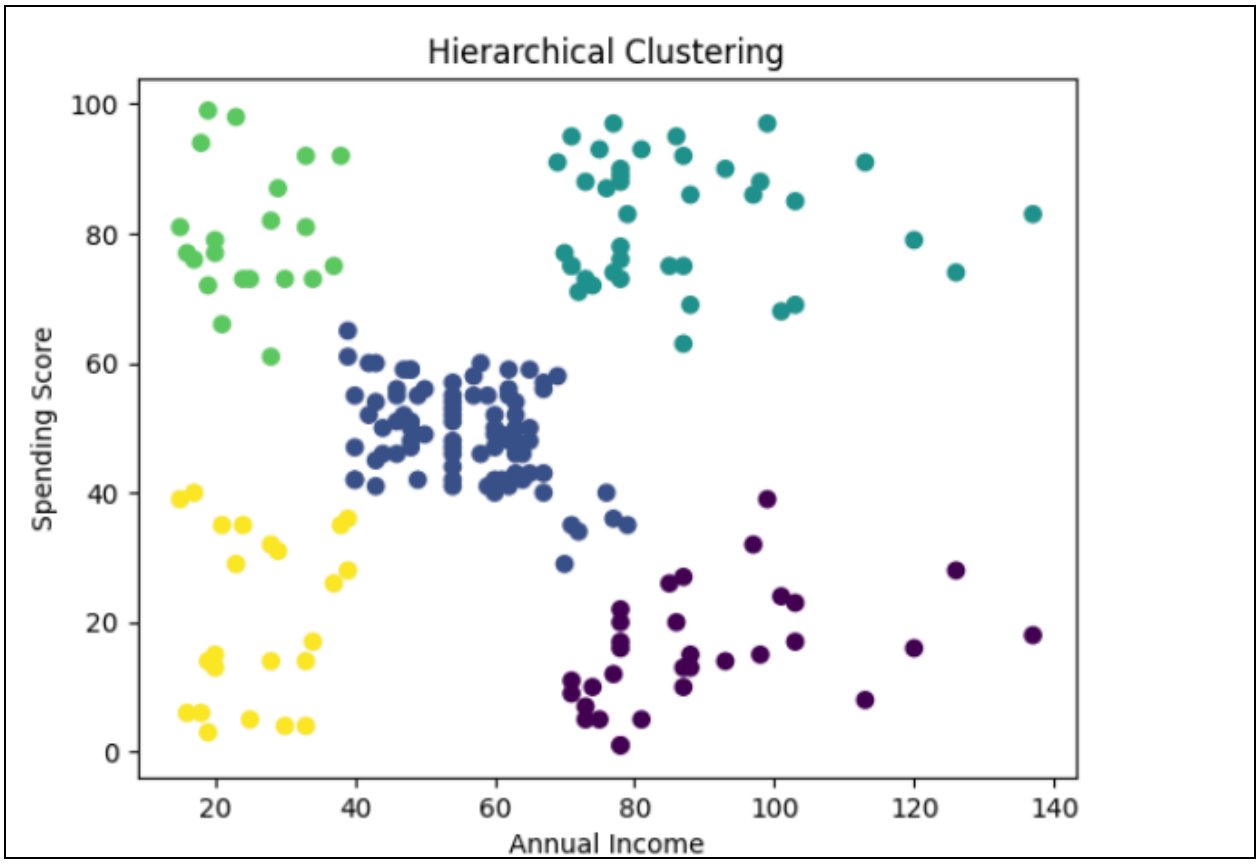
```
hc = AgglomerativeClustering(n_clusters=5)

y_hc = hc.fit_predict(X)
```

✓ 0.0s

Step 9: Visualize Hierarchical Clusters

```
plt.scatter(X['Annual Income (k$)'], X['Spending Score (1-100)'], c=y_hc)
plt.xlabel("Annual Income")
plt.ylabel("Spending Score")
plt.title("Hierarchical Clustering")
plt.show()
```

✓ 0.3s



Conclusion

In this experiment, **K-Means and Hierarchical Clustering algorithms** were successfully applied to the customer segmentation dataset. The algorithms grouped customers based on income and spending patterns, helping identify distinct customer segments. These clustering techniques are useful for marketing analysis, customer targeting, and business decision-making.